INTERNATIONAL BACCALAUREATE

Mathematics: analysis and approaches

MAA

EXERCISES [MAA 2.2]

QUADRATICS

O. Practice questions

1. [Maximum mark: 10] ***[without GDC]***

Solve the equations

(a) $x^2 - 3x - 4 = 0$. [4]

(b) $x^2 - 3x = 0$. [3]

(c) $x^2 - 4 = 0$. [3]

2. [Maximum mark: 15] ***[without GDC]***

Let $f(x) = x^2 - 2x - 3$ and $g(x) = 2x^2 - 4x - 6$ [that is $g(x) = 2f(x)$]

(a) Find the discriminant of f and **hence** solve the equation $f(x) = 0$. [4]

(b) Find the discriminant of g and **hence** solve the equation $g(x) = 0$. [4]

(c) Find the vertex of each function. [4]

(d) Sketch the graphs of the two functions on the same diagram. [3]

3. [Maximum mark: 33] ***[with GDC]***

Complete the following table for the three quadratic functions below

	$f(x) = 2x^2 - 12x + 10$	$f(x) = 2x^2 - 12x + 18$	$f(x) = 2x^2 - 12x + 23$
Discriminant			
y-intercept			
Roots			
Factorisation			
axis of symmetry			
Vertex			
Vertex form			
Solve $f(x) \geq 0$			
Solve $f(x) > 0$			
Solve $f(x) \leq 0$			
Solve $f(x) < 0$			

4. [Maximum mark: 12] ***[with GDC]***

Consider the quadratic $y = 4x^2 - 120x + 800$

(a) (i) Find the roots.

(ii) **Hence** express the quadratic in the form $y = a(x - x_1)(x - x_2)$ [3]

(b) (i) Find the coordinates of the vertex.

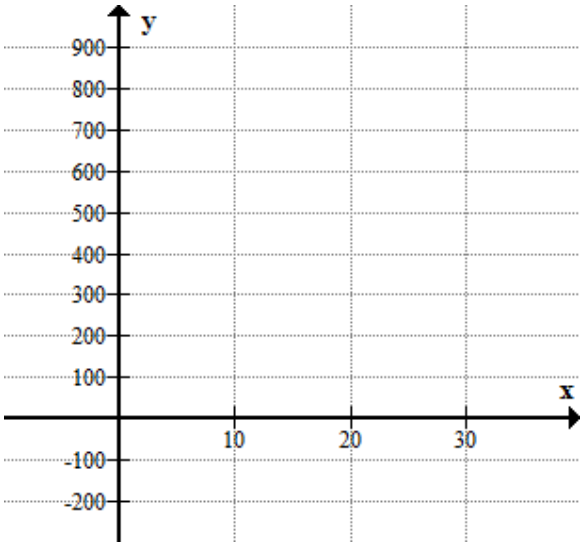
(ii) **Hence** express the quadratic in the form $y = a(x - h)^2 + k$

(iii) Write down the equation of the axis of symmetry

(iv) Write down the minimum value of y [5]

(c) Write down the y - intercept of the quadratic [1]

(d) Draw the graph of the quadratic on the diagram below. [3]



5. [Maximum mark: 14] ***[with GDC]***

Consider the quadratic $y = -4x^2 + 120x - 800$

(a) (i) Find the roots.

(ii) **Hence** express the quadratic in the form $y = a(x - x_1)(x - x_2)$ [3]

(b) (i) Find the coordinates of the vertex.

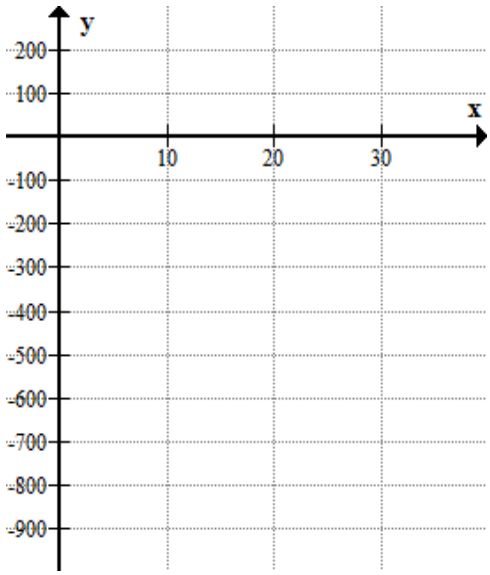
(ii) **Hence** express the quadratic in the form $y = a(x - h)^2 + k$

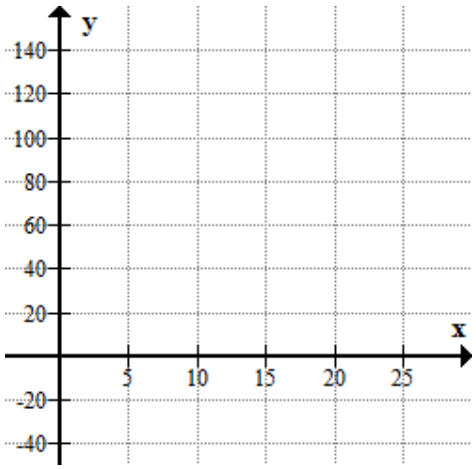
(iii) Write down the equation of the axis of symmetry

(iv) Write down the maximum value of y [5]

(c) Write down the y - intercept of the quadratic [1]

(d) Draw the graph of the quadratic on each of the diagrams below. [5]



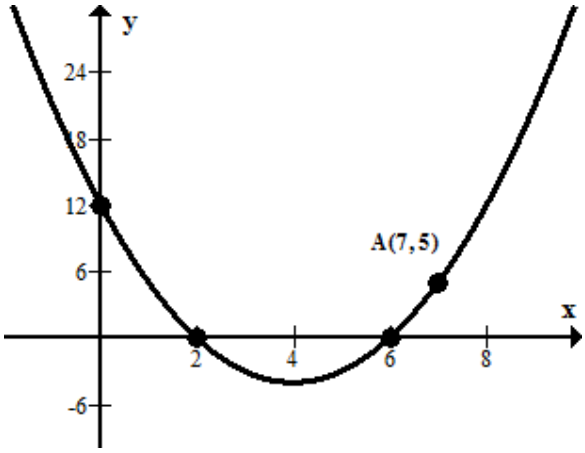


6. [Maximum mark: 8] ***[without GDC]***

The parabola of a quadratic $y = f(x)$ is shown below.

The x-intercepts are $x = 2$ and $x = 6$.

The y-intercept is $y = 12$. The curve passes through the point A(7,5).



(a) By using the symmetry of the graph

(i) Write down the equation of the axis of symmetry.

(ii) Find the value of $f(8)$. Justify your answer.

(iii) Find the value of $f(1)$. Justify your answer. [5]

(b) Find the equation of the curve in the form $f(x) = ax^2 + bx + c$. [3]

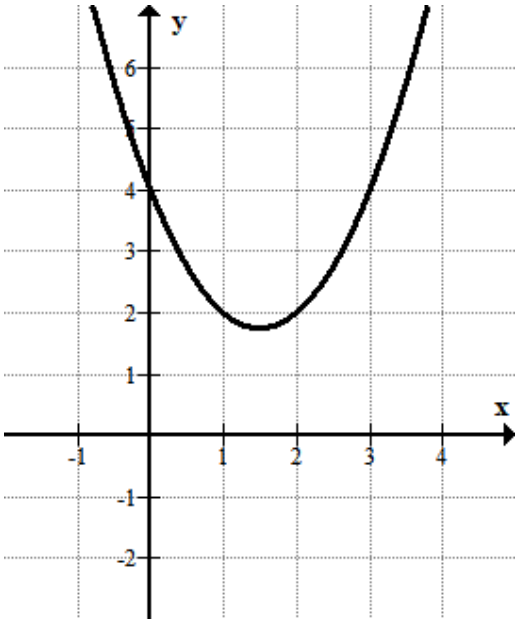
7*. [Maximum mark: 12] ***[with / without GDC]***

Consider the quadratic $y = x^2 - 3x + 4$.

(a) Find the points of intersection of its curve with each of the following lines.

(i) $y = x + 1$ (ii) $y = x$ (iii) $y = x - 1$ [9]

(b) The graph of the quadratic is shown on the diagram below. On the same diagram represent the results of question (a). [3]



A. Exam style questions (SHORT)

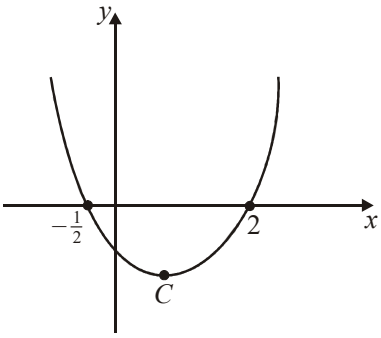
8. [Maximum mark: 4] ***[with / without GDC]***

(b) Solve the equation $x^2 - 3x - 10 = 0$. [2]

(a) Factorize $x^2 - 3x - 10$. [2]

9. [Maximum mark: 4] ***[without GDC]***

The diagram represents the graph of the function $f : x \mapsto (x - p)(x - q)$.

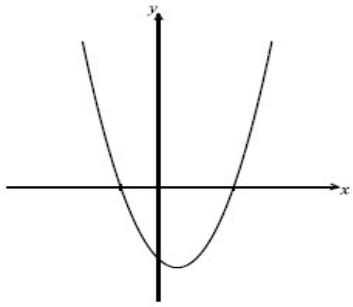


(a) Write down the values of p and q. [2]

(b) The function has a minimum value at the point C. Find the x-coordinate of C. [2]

10. [Maximum mark: 6] ***[without GDC]***

The following diagram shows part of the graph of f, where $f(x) = x^2 - x - 2$.

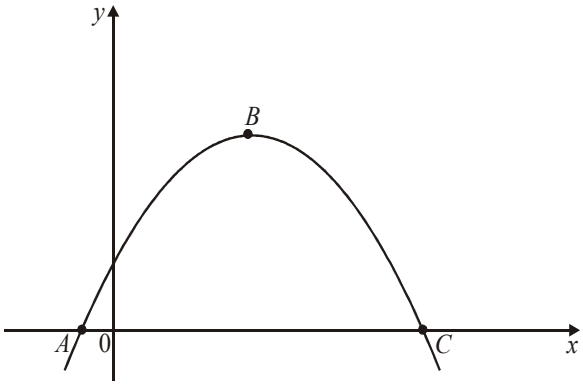


(a) Find both x-intercepts. [4]

(b) Find the x-coordinate of the vertex. [2]

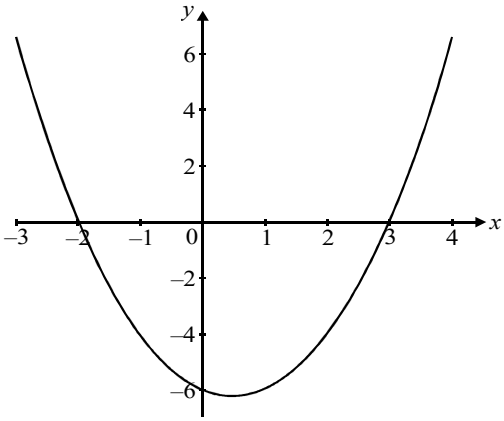
11. [Maximum mark: 4] ***[without GDC]***

The diagram shows the parabola $y = (7 - x)(1 + x)$. The point B is the maximum point and the points A, C are the x-intercepts. Find the coordinates of A, B and C.



12. [Maximum mark: 4] ***[without GDC]***

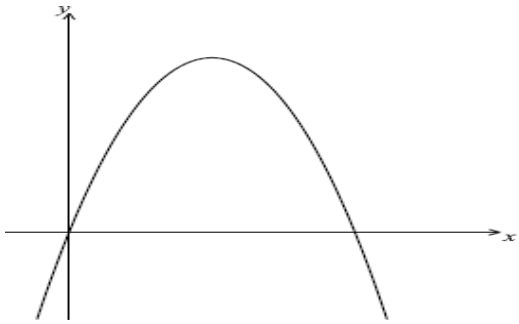
The diagram shows part of the graph with equation $y = x^2 + px + q$. The graph cuts the x- axis at -2 and 3.



Find the value of (i) p; (ii) q.

13. [Maximum mark: 6] ***[without GDC]***

Let $f(x) = 8x - 2x^2$. Part of the graph of f is shown below.



(a) Find the x-intercepts of the graph. [3]

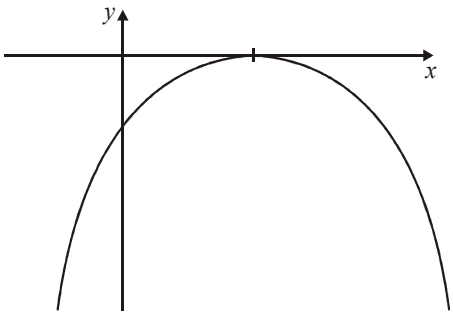
(b) (i) Write down the equation of the axis of symmetry.

(ii) Find the y-coordinate of the vertex. [3]

14. [Maximum mark: 5] ***[without GDC]***

The following diagram shows the graph of function $y = ax^2 + bx + c$.

Complete the table next to the graph to show whether each expression is positive (+), negative (–) or zero (0).

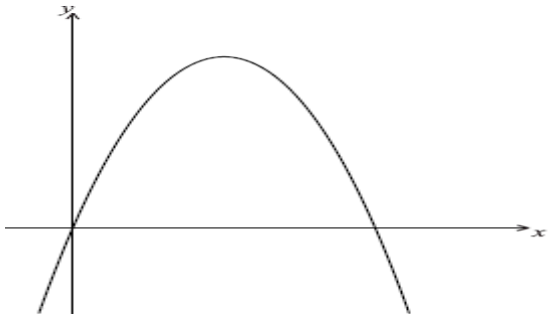


Expression	+ – 0
a	
c	
$b^2 - 4ac$	
$-\frac{b}{2a}$	
b	

15. [Maximum mark: 5] ***[without GDC]***

The following diagram shows the graph of function $y = ax^2 + bx + c$.

Complete the table next to the graph to show whether each expression is positive (+), negative (–) or zero (0).

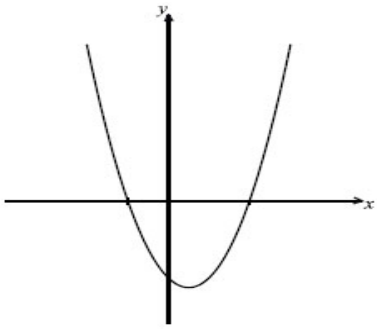


Expression	+ – 0
a	
c	
$b^2 - 4ac$	
$-\frac{b}{2a}$	
b	

16. [Maximum mark: 5] ***[without GDC]***

The following diagram shows the graph of function $y = ax^2 + bx + c$.

Complete the table next to the graph to show whether each expression is positive (+), negative (–) or zero (0).



Expression	+ – 0
a	
c	
$b^2 - 4ac$	
$-\dfrac{b}{2a}$	
b	

17. [Maximum mark: 4] ***[with / without GDC]***

(a) Find the vertex of $f(x) = x^2 - 6x + 14$. [2]

(b) Express the function in the form $f(x) = (x-h)^2 + k$. [2]

18. [Maximum mark: 4] ***[with / without GDC]***

Consider the function $f(x) = 2x^2 - 8x + 5$.

(a) Express $f(x)$ in the form $a(x-p)^2 + q$, where $a, p, q \in \mathbb{Z}$. [3]

(b) Find the minimum value of $f(x)$. [1]

19. [Maximum mark: 4] ***[with / without GDC]***

(a) Find the vertex of $f(x) = 2x^2 + 2x + 2$. [2]

(b) Express the function in the form $f(x) = a(x-h)^2 + k$. [2]

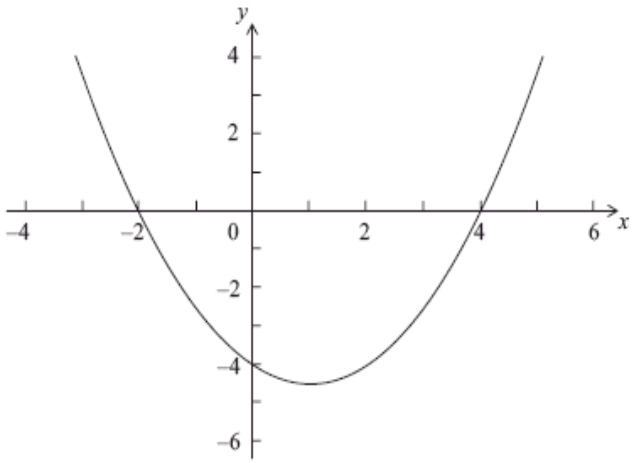
20. [Maximum mark: 4] ***[with / without GDC]***

(a) Find the vertex of $f(x) = -x^2 - x - 1$. [2]

(b) Express the function in the form $f(x) = a(x-h)^2 + k$. [2]

21. [Maximum mark: 6] ***[without GDC]***

Let $f(x) = p(x-q)(x-r)$. Part of the graph of f is shown below.



The graph passes through the points (–2, 0), (0, –4) and (4, 0).

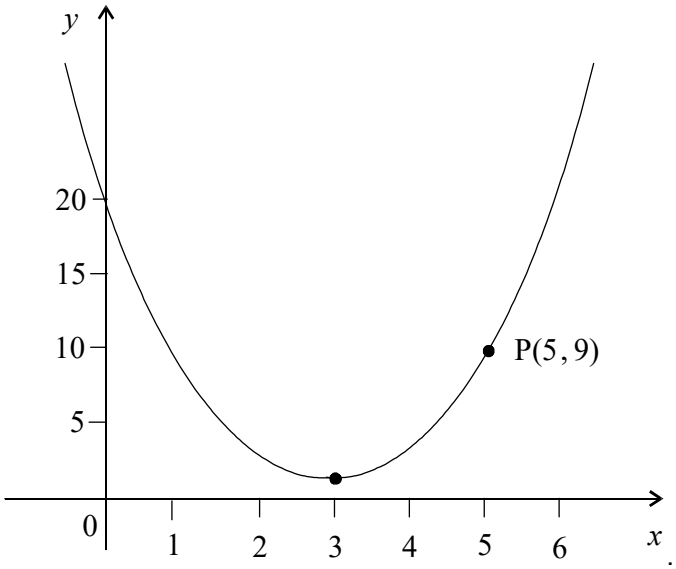
(a) Write down the value of q and of r. [2]

(b) Write down the equation of the axis of symmetry. [1]

(c) Find the value of p. [3]

22. [Maximum mark: 6] ***[without GDC]***

The diagram shows part of the graph of the curve $y = a(x-h)^2 + k$, where $a, h, k \in \mathbb{Z}$.



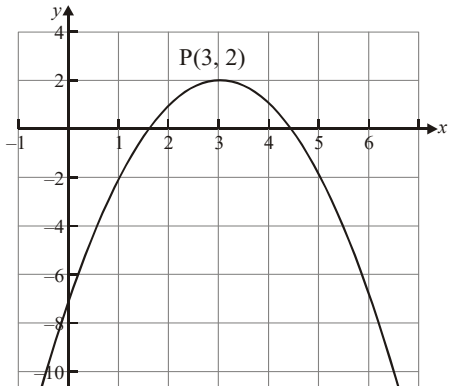
(a) The vertex is at the point $(3, 1)$. Write down the value of h and of k. [2]

(b) The point $\mathrm{P}\ (5, 9)$ is on the graph. Show that $a = 2$. [3]

(c) Hence show that the equation of the curve can be written as $y = 2x^2 - 12x + 19$. [1]

23. [Maximum mark: 5] ***[without GDC]***

The function $f(x)$ is defined as $f(x) = -(x-h)^2 + k$. The diagram below shows part of the graph of $f(x)$. The maximum point on the curve is $\mathrm{P}\ (3, 2)$.



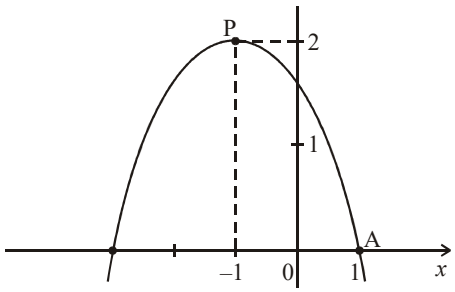
(a) Write down the value of (i) h (ii) k. [2]

(b) Write down the possible values of $f(x)$. [2]

(c) Show that $f(x)$ can be written as $f(x) = -x^2 + 6x - 7$. [1]

24. [Maximum mark: 6] ***[without GDC]***

The diagram shows part of the graph of $y = a\,(x-h)^2 + k$. The graph has its vertex at P, and passes through the point A with coordinates $(1, 0)$.

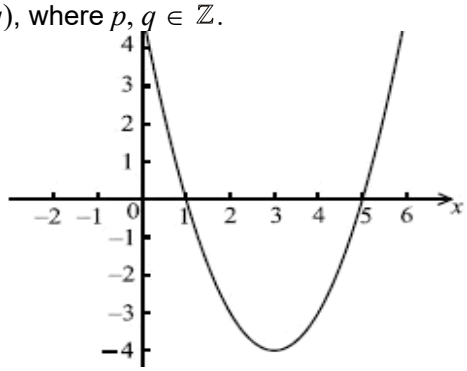


(a) Write down the value of (i) h (ii) k [2]

(b) Calculate the value of a. [3]

25. [Maximum mark: 6] ***[without GDC]***

The following diagram shows part of the graph of a quadratic function, with equation in the form $y = (x-p)(x-q)$, where $p, q \in \mathbb{Z}$.



(a) (i) Write down the value of p and of q

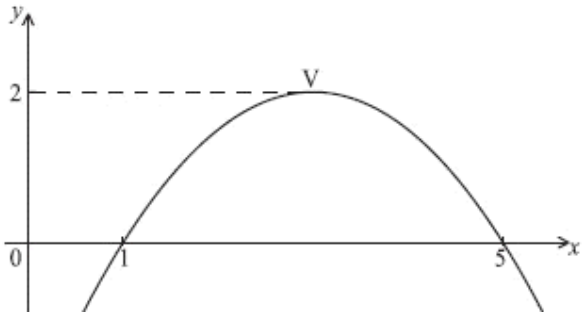
(ii) Write down the equation of the axis of symmetry of the curve [3]

(b) Find the equation of the function in the form $y = (x-h)^2 + k$, where $h, k \in \mathbb{Z}$. [3]

26. [Maximum mark: 5] ***[without GDC]***

Part of the graph of the function $y = d\,(x - m)^2 + p$ is given in the diagram below.

The x-intercepts are $(1, 0)$ and $(5, 0)$. The vertex is $\mathrm{V}(m, 2)$.

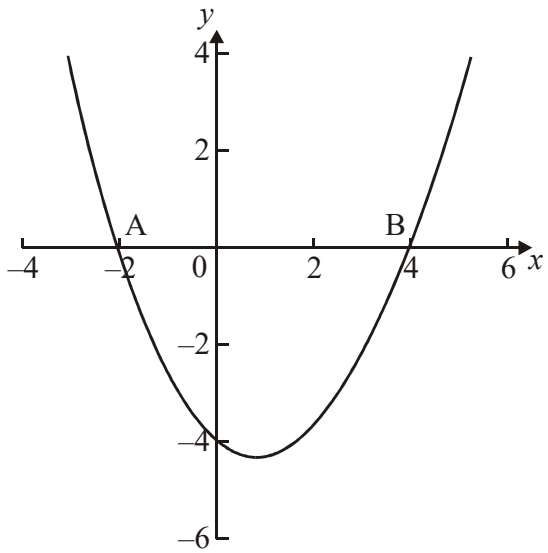


(a) Write down the value of (i) m (ii) p [3]

(b) Find d. [2]

27. [Maximum mark: 8] ***[with / without GDC]***

The equation of a curve may be written in the form $y = a(x - p)(x - q)$. The curve intersects the x-axis at $\mathrm{A}(-2, 0)$ and $\mathrm{B}(4, 0)$. The curve of $y = f(x)$ is shown in the diagram below.



(a) Write down the value of p and of q. [2]

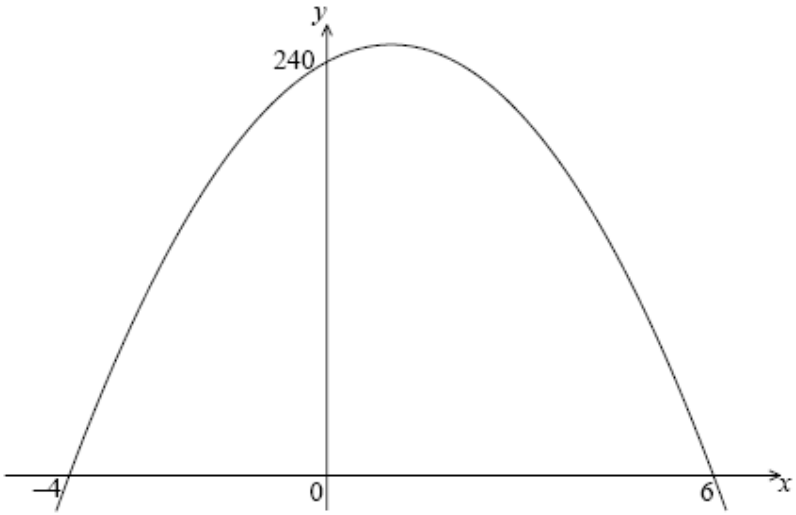
(b) Given that the point $(6, 8)$ is on the curve, find the value of a. [2]

(c) Write the equation of the curve in the form $y = ax^2 + bx + c$. [1]

(d) Write the equation of the curve in the form $y = a\,(x - h)^2 + k$, where $h, k \in \mathbb{Z}$.. [3]

28. [Maximum mark: 8] ***[without GDC]***

The following diagram shows part of the graph of a quadratic function f.



The x-intercepts are at $(-4, 0)$ and $(6, 0)$ and the y-intercept is at $(0, 240)$.

(a) Write down $f(x)$ in the form $f(x) = -10(x-p)(x-q)$. [2]

(b) Find another expression for $f(x)$ in the form $f(x) = -10(x-h)^2 + k$. [4]

(c) Show that $f(x)$ can also be written in the form $f(x) = 240 + 20x - 10x^2$. [2]

29. [Maximum mark: 6] ***[with / without GDC]***

Find the points of intersection between $y = x^2 - 5x + 3$ and $y = 3x - 9$

and sketch a graph to demonstrate the result.

30. [Maximum mark: 6] ***[with / without GDC]***

Find the point of intersection between $y = x^2 - 5x + 3$ and $y = 3x - 13$

and sketch a graph to demonstrate the result.

31. [Maximum mark: 6] ***[with / without GDC]***

Find the points of intersection (if any) between $y = x^2 - 5x + 3$ and $y = 3x - 15$

and sketch a graph to demonstrate the result.

32. [Maximum mark: 6] ***[with / without GDC]***

Find the points of intersection (if any) between $y = x^2 - 3$ and $y = 5 - x^2$

and sketch a graph to demonstrate the result.

33*. [Maximum mark: 4] ***[with / without GDC]***

The quadratic equation $4x^2 + 4kx + 9 = 0$, $k > 0$ has exactly one solution for x.

Find the value of k.

34*. [Maximum mark: 6] ***[with / without GDC]***

The equation $kx^2 + 3x + 1 = 0$ has exactly one solution. Find the value of k.

35*. [Maximum mark: 7] ***[without GDC]***

The quadratic equation $kx^2 + (k-3)x + 1 = 0$ has two equal real roots.

(a) Find the possible values of k. [5]

(b) Write down the values of k for which $x^2 + (k-3)x + k = 0$ has two equal real roots. [2]

36*. [Maximum mark: 7] ***[with GDC]***

Consider $f(x) = 2kx^2 - 4kx + 1 = 0$, for $k \neq 0$. The equation $f(x) = 0$ has two equal roots.

(a) Find the value of k. [5]

(b) The line $y = p$ intersects the graph of f. Find all possible values of p. [2]

37*. [Maximum mark: 6] ***[with / without GDC]***

The equation $x^2 - 2kx + 1 = 0$ has two distinct real roots. Find the set of all possible values of k.

38*. [Maximum mark: 6] ***[with / without GDC]***

A family of functions is given by $f(x) = x^2 + 3x + k$, where $k \in \{1, 2, 3, 4, 5, 6, 7\}$.

Find the possible values of k if the curve of this function crosses the x-axis.

39*. [Maximum mark: 6] ***[with GDC]***

The equation $kx^2 - 3x + (k+2) = 0$ has two distinct real roots. Find the set of possible values of k.

40*. [Maximum mark: 6] ***[with GDC]***

The function f is defined by $f(x) = x^2 - 2x + k(3k+2)$ where $k \in \mathbb{R}$. Find the set of values of k for which $f(x) = 0$ has two distinct real roots.

41*. [Maximum mark: 6] ***[with GDC]***

Consider the equation $(1+2k)x^2 - 10x + k - 2 = 0$, $k \in \mathbb{R}$. Find the set of values of k for which the equation has real roots.

42*. [Maximum mark: 6] ***[with GDC]***

The quadratic function Q is defined by $Q(x) = kx^2 - (k-3)x + (k-8)$, $k \in \mathbb{R}$.

Determine the values of k for which $Q(x) = 0$ has no real roots.

43.** [Maximum mark: 6] ***[with / without GDC]***

Given $f(x) = x^2 + x(2-k) + k^2$, find the range of values of k for which $f(x) > 0$ for all real values of x.

44.** [Maximum mark: 6] ***[without GDC]***

Find the range of values of m such that for all x, $m(x+1) \le x^2$.

45*. [Maximum mark: 5] ***[without GDC]***

For what values of m is the line $y = mx + 5$ a tangent to the parabola $y = 4 - x^2$?

46.** [Maximum mark: 5] ***[without GDC]***

The parabolas $y = 2x^2 + 2x - 1$ and $y = x^2 - m$ have only one common point (they are tangent to each other). Find the value of m.

B. Exam style questions (LONG)

47. [Maximum mark: 12] ***[without GDC]***

Let $f(x) = a(x-4)^2 + 8$.

(a) Write down the coordinates of the vertex of the curve of f. [2]

(b) Given that $f(7) = -10$, find the value of a. [2]

(c) Hence find the y-intercept of the curve of f. [2]

(d) Find the x-intercepts of the curve of f. [4]

(e) Sketch the graph of the curve of f. [2]

48*. [Maximum mark: 11] ***[with GDC]***

Let $f(x) = ax^2 + bx + c$ where a, b and c are rational numbers.

(a) The point $\mathrm{P}(-4, 3)$ lies on the curve of f. Show that $16a - 4b + c = 3$. [2]

(b) The points $\mathrm{Q}(6, 3)$ and $\mathrm{R}(-2, -1)$ also lie on the curve of f. Write down two other linear equations in a, b and c. [4]

(c) **Hence** find $f(x)$. [2]

(d) Write $f(x)$ in the form $f(x) = a(x-h)^2 + k$, where a, h and k are rational numbers. [3]

49*. [Maximum mark: 10] ***[with / without GDC]***

Consider two different quadratic functions of the form $f(x) = 4x^2 - qx + 25$. The graph of each function has its vertex on the x-axis.

(a) Find both values of q. [3]

(b) For the greater value of q, solve $f(x) = 0$. [2]

(c) Find the coordinates of the point of intersection of the two graphs. [2]

(d) Skecth the graphs of the two functions on the same diagram. [3]

50*. [Maximum mark: 12] ***[without GDC]***

Let $f(x) = 3x^2 - x + 4$.

(a) Find the values of m for which the line $y = mx + 1$

(i) is a tangent to the graph of f.

(ii) meets the graph of f.

(iii) does not meet the graph of f. [6]

(b) For each value of m found in question (a)(i), find the common point of the two graphs. [6]

51.** [Maximum mark: 10] ***[with GDC]***

Consider the graph *G* of the function $f(x) = x^2$ and the point M(5,2) on the Cartesian plane.

(a) Find the coordinates of a point A on *G* and a point B on *x*-axis, given that M is the midpoint of the line segment [AB]. [5]

(b) Find the coordinates of a point C on *G*, given that the distance between M and C is 23. [5]